

Pape DIAWARA

Software Engineer – Performance Analytics & Data Platforms

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Summary

Software Engineer specializing in Performance Analytics, data platforms, and backend engineering. Experienced in building scalable Python services, analytics platforms, and internal developer tools that enable engineering teams to measure application performance, identify performance regressions, and make data-driven decisions. Strong expertise in Python, TypeScript, large-scale data processing, and building production software that transforms complex telemetry into meaningful operational insights.

Programming

Skills

Languages: Python, TypeScript, JavaScript, SQL

Performance Analytics: Performance Monitoring, Performance Benchmarking, Experimentation, Telemetry Analysis, Root Cause Analysis, Data Visualization

Software Engineering: FastAPI, Django, REST APIs, AsyncIO, React

Data Engineering: Pandas, Polars, NumPy, ETL Pipelines, Data Modeling, Data Validation, Data Quality

Infrastructure: AWS, Docker, Kubernetes, Helm, PostgreSQL, Redis, GitLab CI/CD, Azure DevOps

Education

EPITA - School of Engineering and Computer Science, PARIS	Computer Engineering Degree • Master
Computer Science	
https://www.epita.fr/en/homepage/	

Experience

BNP Paribas Asset Management	Paris
Software Engineer – Performance Analytics & Data Platforms	February 2025 - Present
- Designed and built scalable analytics platforms and Python services supporting portfolio analytics and engineering workflows across multiple investment teams. - Developed high-performance data pipelines processing millions of financial records across 15+ data sources, improving execution time by 35% and memory efficiency through vectorized processing using Pandas, NumPy, and Polars. - Built internal analytics platforms and reusable APIs enabling engineering and analytics teams to explore performance metrics, investigate performance anomalies, and independently validate software releases. - Designed engineering analytics workflows combining statistical methods, experimentation, and regression detection to deliver operational insights and improve production reliability. - Optimized backend services through asynchronous processing, query optimization, caching, and profiling, improving application throughput by 2x while reducing response times. - Developed production-grade software with automated testing, monitoring, observability, and CI/CD practices, maintaining 99.9% production reliability across multiple environments. - Led technical initiatives, conducted code reviews, and mentored engineers while promoting software engineering best practices across the team.	
SEGULA TECHNOLOGIE	Paris
Software Engineer – Data & Performance Analytics	2022-11-29 - 2024-10-30
- Designed and developed analytics platforms supporting predictive maintenance, semantic search, and intelligent engineering applications. - Built scalable backend services processing millions of telemetry events used by engineering and research teams. - Developed engineering analytics platforms enabling engineers to monitor software performance, compare software releases, and investigate performance degradations through intuitive web interfaces. - Designed large-scale ETL pipelines integrating data from 10+ heterogeneous sources supporting experimentation, statistical analysis, and machine learning workflows. - Implemented performance monitoring, regression detection, and automated validation processes improving the reliability of engineering analytics. - Developed interactive web applications using React, TypeScript, FastAPI, and Django to expose analytics through intuitive user interfaces. - Improved backend throughput by 2x through asynchronous processing, SQL optimization, Redis caching, and performance profiling. - Integrated AI-assisted classification and semantic search capabilities improving data discovery and engineering productivity.	
GAWL, Paris	2022-09-01 - 2022-11-24
Software Engineer – Analytics Platform	
- Built a real-time analytics platform collecting and visualizing operational and application performance metrics. - Designed ETL pipelines processing hundreds of thousands of operational events daily for reporting and engineering analysis. - Developed interactive analytics dashboards and backend services enabling engineering teams to independently explore operational metrics, software behavior, and application performance. - Implemented monitoring, validation, alerting, and data-quality controls improving production reliability. - Optimized backend services, SQL queries, and caching mechanisms, reducing dashboard response time by 45%.	
Freelance	2020 - 2022
Software Engineer	
- Designed and developed custom backend systems, analytics platforms, and internal engineering tools for SMEs. - Built production-grade web applications automating operational workflows and improving data accessibility. - Developed scalable APIs, reporting platforms, and analytics dashboards supporting business decision-making. - Delivered end-to-end software solutions from architecture and implementation through deployment, optimization, and long-term maintenance.	
SEDIMA GROUP	2017-11-01 - 2019-08-31
Software Engineer	
- Designed and developed internal business applications supporting order management, inventory, payments, and operational workflows. - Built reporting tools and analytics features improving operational visibility and business decision-making. - Automated business processes through scalable backend services and structured data workflows. - Improved application reliability, maintainability, and overall software quality across multiple internal platforms.	
Emc2 Group	2017-07-03 - 2017-08-31
Backend Software Engineer	
- Developed backend services and internal web applications using Python and Django. - Designed reusable software components, REST APIs, and structured data models. - Implemented automated testing and contributed to maintaining production-ready software. - Collaborated with cross-functional teams to deliver reliable and maintainable backend solutions.	